

**CSE 370– Database Systems**

**QUIZ 4 (Set B)**

**Marks**

**Full Name (in Block Letter):** \_\_\_\_\_

**ID:** \_\_\_\_\_ **Section:** \_\_\_\_ **Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_ **Total Marks: 20** **Time: 15 minutes**

**Question 1:**

| File 1 |  | ID | Name     | Price |   | File 2 |
|--------|--|----|----------|-------|---|--------|
| 1      |  | 6  | Carrot   | 30    | ← | 30     |
| 2      |  | 2  | Tomato   | 50    |   | 80     |
| 3      |  | 10 | Spinach  | 70    |   | 150    |
| 4      |  | 4  | Broccoli | 80    | ← | 250    |
| 5      |  | 3  | Capsicum | 100   |   |        |
| 6      |  | 9  | Cabbage  | 120   |   |        |
| 7      |  | 5  | Radish   | 150   | ← |        |
| 8      |  | 8  | Eggplant | 180   |   |        |
| 9      |  | 1  | Peas     | 200   |   |        |
| 10     |  | 7  | Mushroom | 250   | ← |        |

The **Name** column is selected as a new index. **Construct** a B+ tree of order  $n = 3$  with the **Name** column as the search key value. The search key values are inserted one by one in the same order as shown in the above table. While inserting in a tree, alphabetical order is maintained, if two vegetable names start with the same letters, second characters are compared and so on.

**For each insertion, a new B+ tree must be drawn.**



